In the 4h version.

3.29* Discuss the limiting effect of repeatedly filtering an image with a 3×3 lowpass filter kernel. You may ignore border effects.

3.31* An image is filtered with a kernel whose coefficients sum to 1. Show that the sum of the pixel values in the original and filtered images is the same.

3.37 Do the following:

(a)* Develop a procedure for computing the median of an $n \times n$ neighborhood.

(b) Propose a technique for updating the median as the center of the neighborhood is moved from pixel to pixel.

3.40* You saw in Fig. 3.46 that the Laplacian with a -8 in the center yields sharper results than the one with a -4 in the center. Explain the reason why.

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结论：忽略边界效应时反复用 3×3 的均值滤波，结果会越来越模糊，最终整幅图像收敛为常数图，其值等于原图的全局平均灰度。

简要解释：设图像为 $f(x,y)$，以核 $h=\begin{bmatrix}\frac{1}{9}&\frac{1}{9}&\frac{1}{9}\\ \frac{1}{9}&\frac{1}{9}&\frac{1}{9}\\ \frac{1}{9}&\frac{1}{9}&\frac{1}{9}\end{bmatrix}$ 为例，

得到 $g=f*h$，连续做 k 次就是 $g^{(k)}=f*h^{(*k)}$

$h(x,y)$ 的二维 DTFT 为：

$$H(\omega_x,\omega_y)=\sum_x\sum_y h(x,y)e^{-j(x\omega_x+y\omega_y)}$$

$$=\frac{1}{9}\sum_{x=-1}^{1}\sum_{y=-1}^{1}e^{-j(x\omega_x+y\omega_y)}=\frac{1}{9}\left(\sum_{x=-1}^{1}e^{-jx\omega_x}\right)\left(\sum_{y=-1}^{1}e^{-jy\omega_y}\right)$$

其中 $\sum_{x=-1}^{1}e^{-jx\omega_x}=e^{j\omega_x}+1+e^{-j\omega_x}=1+2\cos\omega_x$

同理，$\sum_{y=-1}^{1}e^{-jy\omega_y}=1+2\cos\omega_y$

所以 $H(\omega_x,\omega_y)=\frac{1}{9}(1+2\cos\omega_x)(1+2\cos\omega_y)$

做 k 次卷积就是 H^k。除了 $(\omega_x,\omega_y)=(0,0)$，那直流分量外，$|H|<1$，所以所有非零频率都会被 $|H|^k$ 几何衰减到 0，只剩直流分量，也就是整幅图的平均值。

3.31* An image is filtered with a kernel whose coefficients sum to 1. Show that the sum of the pixel values in the original and filtered images is the same.

设 $g=f*h$，则 $\sum_{x,y}g(x,y)=\sum_{x,y}\sum_{i,j}h(i,j)f(x-i,y-j)$

$$=\sum_{i,j}h(i,j)\cdot\sum_{x,y}f(x-i,y-j)$$

平移不改变求和，有 $\sum_{x,y}f(x-i,y-j)=\sum_{x,y}f(x,y)$

于是 $\sum_{x,y}g(x,y)=\left[\sum_{i,j}h(i,j)\right]\cdot\left[\sum_{x,y}f(x,y)\right]$，当 $\sum_{i,j}h(i,j)=1$ 时 $\sum_{x,y}g(x,y)=\sum_{x,y}f(x,y)$

3.37 Do the following:

(a)* Develop a procedure for computing the median of an $n \times n$ neighborhood.

(b) Propose a technique for updating the median as the center of the neighborhood is moved from pixel to pixel.

利用直方图

(a) ① 构建一个灰度直方图 H[0, ··· L-1], 统计窗口内各灰度级在窗口内出现次数

② 累加直方图, 找到最小灰度值 m, 使得 $\sum_{i=0}^{m} H[i] \geq \frac{n^2}{2}$

则 m 就是中值

(b) 使用直方图进行增量更新

① 维护当前窗口的直方图 H

② 当窗口移动时,

对离开的像素值灰度 p_{out} 执行 $H[p_{out}]--$

对进入的像素值灰度 p_{in}, 执行 $H[p_{in}]++$

③ 更新中值

在原来的中值附近查找

如果计数小于 $\frac{n^2}{2}$, 向右移动; 如果计数大于 $\frac{n^2}{2}$, 向左移动

因为窗口相邻, 变化不大, 所以中值更新很快

3.40* You saw in Fig. 3.46 that the Laplacian with a −8 in the center yields sharper results than the one with a −4 in the center. Explain the reason why.

四邻域的算子

$\nabla^2 f(x,y) = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ 由锐化公式 $g = f - \nabla^2 f$, 最终的锐化滤波核

$S_4 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

八邻域算子:

$$\nabla^2 f(x,y) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$S_8 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 9 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

锐化的本质就是增强中心像素与周围像素的差异.

S_8 的滤波核给了中心像素权重为9, 远大于 S_4 的5,

所以-8的算子会比-4的算子更强烈的突出中心像素的原始值

当一个物体本身与周围像素差异较大时, 处理的计算结果

会有更大的变化, 从而形成更锐利的边缘